

The Program in Action

When you run *progressbar.py*, the output will look like this:

```
Progress Bar Simulation, by Al Sweigart
[██████████] 24.6% 1007/4098
```

How It Works

The progress bar relies on a certain trick that programs running in terminal windows can perform. Just as `'\n'` and `'\t'` are escape characters for newline and tab characters, respectively, `'\b'` is an escape character for backspace characters. If you “print” a backspace character, the text cursor will move to the left and erase the previously printed character. This only works for the current line the text cursor is on. If you run the code `print('Hello\b\b\b\b\bHowdy')`, Python will print the text “Hello,” move the text cursor back five spaces, and then print the text “Howdy.” The “Howdy” text will overwrite “Hello,” making it look as though you called “Howdy.”

We can use this technique to create an animated progress bar on a single line by printing one version of the bar, moving the text cursor back to the start, then printing an updated progress bar. This effect can generate any text animation without requiring a module such as `bext`, though it will be limited to taking up a single line in the terminal window.

Once you’ve created this program, you can display progress bars in your other Python programs by running `import progressbar` and printing the string returned from `progressbar.getProgressBar()`.

```
1. """Progress Bar Simulation, by Al Sweigart al@inventwithpython.com
2. A sample progress bar animation that can be used in other programs.
3. View this code at https://nostarch.com/big-book-small-python-projects
4. Tags: tiny, module"""
5.
6. import random, time
7.
8. BAR = chr(9608) # Character 9608 is '█'
9.
10. def main():
11.     # Simulate a download:
12.     print('Progress Bar Simulation, by Al Sweigart')
13.     bytesDownloaded = 0
14.     downloadSize = 4096
15.     while bytesDownloaded < downloadSize:
16.         # "Download" a random amount of "bytes":
17.         bytesDownloaded += random.randint(0, 100)
18.
19.         # Get the progress bar string for this amount of progress:
20.         barStr = getProgressBar(bytesDownloaded, downloadSize)
21.
22.         # Don't print a newline at the end, and immediately flush the
23.         # printed string to the screen:
24.         print(barStr, end='', flush=True)
25.
```